

DSA

My Recommended Roadmap

Semester 1 • Arrays • Strings • Stack • Queue • Linked List • Binary Search 150 questions	Semester 3 • Dynamic Programming • Greedy • Backtracking • Advanced Graphs • Mixed Revision 150 more questions (Total: 450)
Semester 2 • Trees • Heap • HashMap • Sliding Window • Graph Basics • Recursion 150 more questions (Total: 300)	Semester 4 No major new topics. Focus on: • Mock coding interviews • Timed contests • Revision of previously solved problems • Solving unseen Medium-level questions

If your goal is specifically **FAANG** plus strong campus placements, I'd prioritize topics in this order:

1. **Arrays & Strings**
2. **HashMap / HashSet**
3. **Binary Search**
4. **Linked List**
5. **Trees & BST**
6. **Sliding Window & Two Pointers**
7. **Graphs (BFS/DFS)**
8. **Heap / Priority Queue**
9. **Greedy**
10. **Dynamic Programming**
11. **Backtracking**

Hard Questions

Target Only 40–70

Don't solve 300 Hard problems. Waste of time for campus preparation.

Examples

Dynamic Programming: Edit Distance, Word Break, Coin Change

Longest Increasing Subsequence

Graphs: Alien Dictionary, Course Schedule II

Backtracking: Sudoku Solver, N Queens

Tier 1 (Must Master)

These topics appear in almost every company.

Topic	Campus	TCS Prime	Difficulty	
Arrays	★★★★★	★★★★★	Easy–Medium	
Strings	★★★★★	★★★★★	Easy–Medium	
HashMap / HashSet	★★★★★	★★★★★	Medium	
Two Pointers	★★★★★	★★★★★	Medium	
Sliding Window	★★★	★★★★★	Medium	
Binary Search	★★★★★	★★★★★	Medium	
Stack	★★★★★	★★★★	Easy–Medium	
Queue	★★★★	★★★	Easy	
Linked List	★★★★★	★★★★★	Medium	
Recursion	★★★	★★★★	Medium	

Tier 2 (Very Important)

Topic	Campus	TCS Prime	Difficulty
Trees	★★★★★	★★★★★	Medium
BST	★★★	★★★★★	Medium
Heap / Priority Queue	★★★	★★★★	Medium
DFS	★★★	★★★★★	Medium
BFS	★★★	★★★★★	Medium
Graph Basics	★★	★★★★★	Medium
Greedy	★★	★★★★	Medium
Backtracking	★★	★★★★	Medium



Tier 3 (Prime Level)

Topic	Campus	TCS Prime	Difficulty
Dynamic Programming	★	★★★★★	Hard
Trie	★	★★★	Hard
Segment Tree	★	★★	Hard
Union Find (DSU)	★	★★★	Medium
Topological Sort	★	★★★★	Medium
Shortest Path	★	★★★★	Medium
MST (Kruskal/Prim)	★	★★★	Medium